

Lab 8: Server Configuration – DHCP, DNS and Web Server in Cisco Packet Tracer

Objectives

- Configure a server in Cisco Packet Tracer to deliver DHCP, DNS, and HTTP (Web) services.
- Understand how DHCP automatically assigns IP addresses to client devices on a network.
- Explore the DNS process of translating domain names into IP addresses.
- Verify that all configured services work correctly by accessing the web server from client PCs.

Theory

Dynamic Host Configuration Protocol (DHCP)

DHCP is a network management protocol that automatically provides client devices with IP addresses, subnet masks, default gateways, and DNS server information. Rather than requiring administrators to configure each device manually, DHCP enables efficient, error-free address management across the network. When enabled on a Cisco Packet Tracer server, DHCP allows connected PCs and other hosts to request IP settings on demand, greatly reducing administrative workload and eliminating the risk of duplicate IP address conflicts.

Domain Name System (DNS)

DNS serves as the translation layer between human-readable domain names (such as `www.example.com`) and the numerical IP addresses that machines use to communicate. Without DNS, users would need to memorize IP addresses to reach network resources. In Cisco Packet Tracer, setting up DNS on a server enables client devices to resolve domain names and locate the correct IP address for the web server, simulating how real-world DNS servers act as a directory or "phonebook" for the internet.

Web Server (HTTP Service)

A web server hosts web content and responds to client requests using the Hypertext Transfer Protocol (HTTP). When a client types a domain name into a browser, DNS resolves it to the server's IP address, and the HTTP service then delivers the requested web page. In Packet Tracer, activating the HTTP service on the server allows PCs to retrieve and display web content, which serves as confirmation that both DHCP and DNS are functioning as expected.

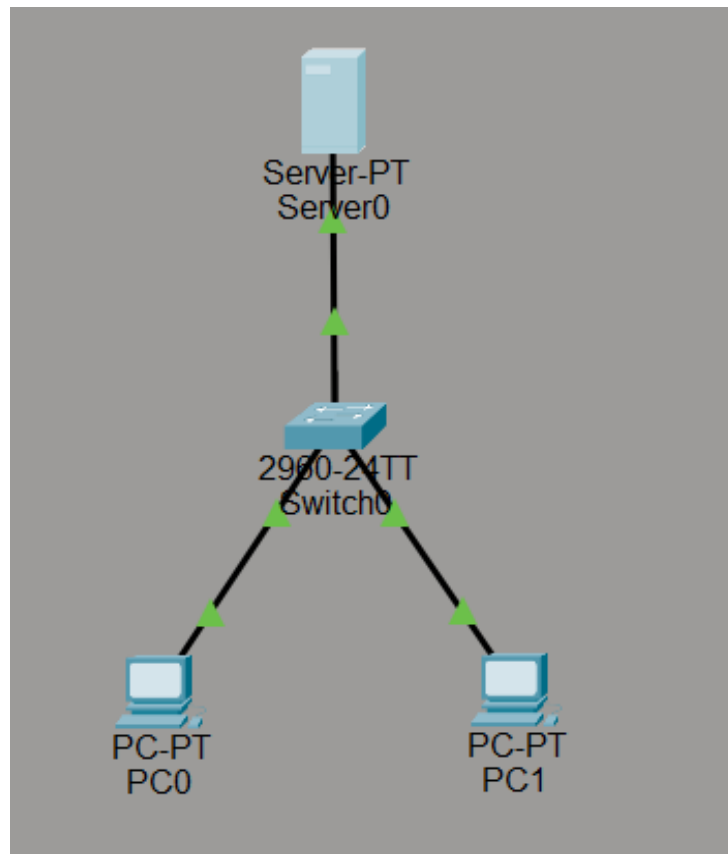
Integration of Services

DHCP, DNS, and HTTP services are designed to work together to deliver seamless network connectivity:

- DHCP handles automatic IP address assignment for all client devices.
- DNS resolves human-friendly domain names to the correct server IP addresses.
- HTTP delivers web content from the server to the requesting client.

Testing these services together within Packet Tracer illustrates how real networks provide automated configuration, name resolution, and web access in a unified, centralized manner.

Network Topology



The network topology used in this lab follows a basic star configuration, with all devices connecting through a central switch. The server acts as the central resource provider, offering DHCP, DNS, and Web services, while two PCs function as client machines for testing connectivity and service functionality. This design demonstrates how a centralized server can efficiently handle IP address management, domain name resolution, and web hosting within a small network environment.

Configuration

IP Addressing Scheme

Device	IP Address	Subnet Mask	Default Gateway
Server	192.168.1.2	255.255.255.0	192.168.1.1
PC	DHCP	DHCP	DHCP

PC Address Details (Dynamically Assigned via DHCP)

Note: The IP addresses listed below are automatically assigned by the DHCP server and do not require manual configuration on the PCs.

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC1	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

Server Configuration Steps

Step 1: Assign a Static IP Address to the Server

Begin by opening the Server device in Packet Tracer. Navigate to Desktop > IP Configuration and enter the following static settings:

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS Server: 192.168.1.2

DHCP Configuration

Step 2: Enable the DHCP Service

On the server, go to Services > DHCP and switch the DHCP service to ON.

Step 3: Create a DHCP Address Pool

Configure the DHCP pool with the following values, then click Add to save the pool:

Field	Value
Pool Name	LANPOOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

Verification

On each client PC, open Desktop > IP Configuration and select the DHCP option. Use the Command Prompt to run a ping test (e.g., ping 192.168.1.1) to confirm connectivity.

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=2ms TTL=128
Reply from 192.168.1.1: bytes=32 time=10ms TTL=128
Reply from 192.168.1.1: bytes=32 time=9ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms
```

DNS Configuration

Step 4: Enable the DNS Service

On the server, navigate to Services > DNS and turn the DNS service ON.

Step 5: Add a DNS Record

Create a new DNS A record with the following details, then click Add:

- Name: www.hcoe.com
- Type: A Record
- Address: 192.168.1.2

Verification

On each client PC, open the Command Prompt and type: ping www.hcoe.com. If DNS is configured correctly, the ping will resolve to 192.168.1.2 and return successful replies.

```
C:\>ping www.hcoe.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Web Server Configuration

Step 6: Enable the HTTP Service

On the server, go to Services > HTTP and turn the HTTP service ON. Edit the index.html file(optional) to customize the web page content displayed to clients.

Verification

On any client PC, open Desktop > Browser and type http://www.hcoe.com into the address bar. The custom web page hosted on the server should load successfully, confirming that DHCP, DNS, and HTTP are all working in coordination.

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>

URL http://www.hcoe.com

Go

Stop

Lab 8

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Network Topology

```
graph TD
    Server[Server-PT Server0] --- Switch[2950-24TT Switch0]
    Switch --- PC0[PC-PT PC0]
    Switch --- PC1[PC-PT PC1]
```

By Sujal Shrestha

Discussion

This lab demonstrated the configuration and integration of three essential network services — DHCP, DNS, and HTTP — within a star topology using Cisco Packet Tracer. The DHCP service effectively automated IP address distribution to client PCs, eliminating the need for manual network configuration. DNS accurately resolved the domain name `www.hcoe.com` to the server's IP address, allowing clients to reach the hosted web page by name rather than by numerical address. The HTTP service then delivered the web content to the requesting clients as expected. Together, these services illustrated how a centralized server simplifies network administration and supports smooth, efficient communication across the network.

Conclusion

The lab successfully confirmed that DHCP, DNS, and Web services can be configured and validated in Cisco Packet Tracer. DHCP streamlined IP address allocation across all client devices, DNS provided reliable domain name resolution, and the web server served content to connected clients. The combined operation of these services highlighted the critical role of centralized server configuration in building efficient, scalable, and user-friendly network environments.